Ch 9.3-4: Support Vector Machine

Lecture 28 - CMSE 381

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Fri, Apr 3, 2026

Announcements

Last time:

- 9.2 Support Vector Classifier

This lecture:

- 9.3 Support Vector Machine

Announcements:

- HW #6 due Sunday 4/12

21	W	3/18	Polynomial & Step Functions	7.1-7.2		
22	F	3/20	Step Functions; Basis functions; Start Splines	7.2-7.4		
23	M	3/23	Regression Splines	7.4		Q7
24	W	3/25	Decision Trees	8.1		
25	F	3/27	Random Forests	8.2.1, 8.2.2	HW #5 Due Sun 3/29	
26	M	3/30	Maximal Margin Classifier	9.1		Q8
27	W	4/1	SVC	9.2		
28	F	4/3	SVM	9.3, 9.4		
29	M	4/6	Single Layer NN	10.1		Q9
30	W	4/8	Multi Layer NN	10.2		
31	F	4/10	CNN	10.3	HW #6 Due Sun 4/12	
32	M	4/13	Unsupervised learning / clustering	12.1, 12.4		Q10
33	W	4/15	Virtual: Project Office Hours			
	F	4/17	***Review***			
	M	4/20	**Midterm #3**			
	W	4/22				
	F	4/24			**Proiect Due**	

Section 1

Last Time

Classification Setup

Data matrix:

$$X = \begin{pmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{pmatrix}_{n \times p}$$

Observations in one of two classes, $y_i \in \{-1, 1\}$

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$$x_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{1p} \end{pmatrix}, \cdots, x_n = \begin{pmatrix} x_{n1} \\ \vdots \\ x_{np} \end{pmatrix}$$

Separate out a test observation

$$x^* = (x_1^* \ \cdots \ x_p^*)^T$$

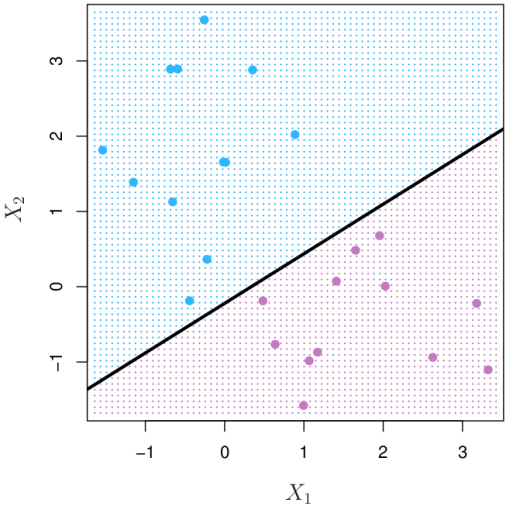
Hyperplane becomes a classifier

If you have a separating hyperplane:

- Check

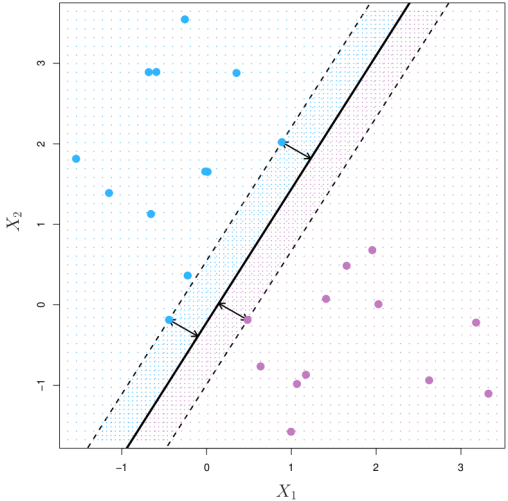
$$f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$$

- If positive, assign $\hat{y} = 1$
- If negative, assign $\hat{y} = -1$



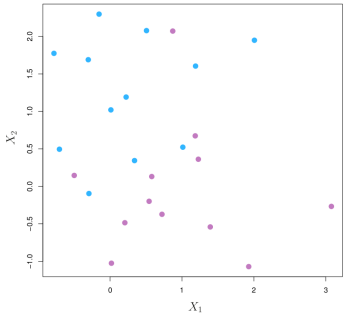
How do we pick? Old version

Maximal margin classifier

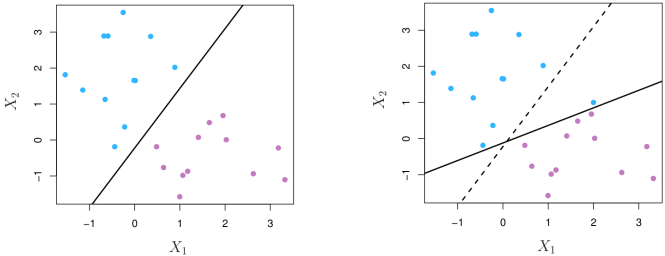


- For a hyperplane, the *margin* is the smallest distance from any data point to the hyperplane.
- Observations that are closest are called *support vectors*.
- The *maximal margin hyperplane* is the hyperplane with the largest margin
- The classifier built from this hyperplane is the *maximal margin classifier*.

Issues

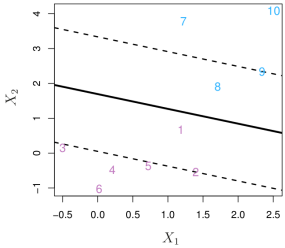


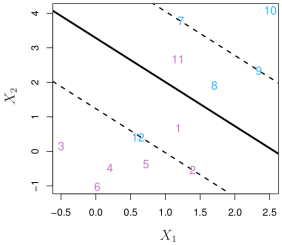
No separating hyperplane exists



Choice of hyperplane is sensitive to new points

Support Vector Classifier





$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

Test your understanding in PollEv!

- Soft margin
- Allow for violations across margin
- Allow for violations across hyperplane

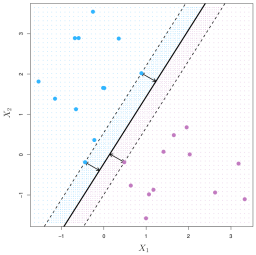
Two formulations side by side

Maximal Margin Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \quad \forall\, i = 1, \ldots, n$$



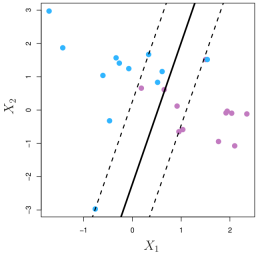
Support Vector Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\, M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$



So many variables

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

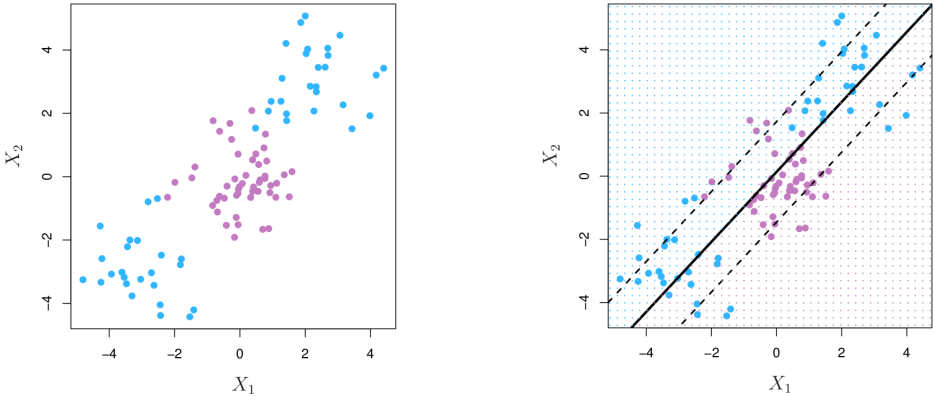
$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

- C is nonnegative tuning parameter
  - Bounds sum of ε_i; number & severity of violating margin (budget)
  - $C = 0$ means no violations allowed
  - $C > 0$ means at most C observations can be on wrong side of hyperplane
- M is the width of the margin
- $\varepsilon_1, \cdots, \varepsilon_n$ are slack variables allowing observations to go to the other side
  - If $\varepsilon_i = 0$, then on correct side of margin
  - If $\varepsilon_i > 0$ then on the wrong side of margin (Violated margin)
  - If $\varepsilon_i > 1$ then on the wrong side of hyperplane

Limiting factor of SVC



- Requires linear boundaries
- "Non-linear" is a lot of things, how do you choose features to learn?
- Right side is result from SVC

What will you learn today?

- How is nonlinearity introduced in Support Vector Machine?
  - You should be able to describe how kernels work conceptually and mathematically.
- What type of kernel makes Support Vector Machine equivalent to Support Vector Classifier?
- What other kernels are there?
- What type of kernel leads to more local behavior?
- How to use Support Vector Machine in Python?
  - You should be able to fit a Support Vector Machine using different kernels covered in class.
  - You should be able select the appropriate hyperparameter that optimize bias-variance trade off.

Section 2

Support Vector Machine

How to bend the plane: polynomial features?

Want $2p$ features:

$$X_1, X_1^2, X_2, X_2^2, \cdots, X_p, X_p^2$$

Optimization becomes:

$$\underset{\beta_0,\beta_{11},\beta_{12},\ldots,\beta_{p1},\beta_{p2},\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to } y_i\left(\beta_0 + \sum_{j=1}^{p}\beta_{j1}x_{ij} + \sum_{j=1}^{p}\beta_{j2}x_{ij}^2\right) \geq M(1-\epsilon_i),$$

$$\sum_{i=1}^{n}\epsilon_i \leq C, \;\; \epsilon_i \geq 0, \;\; \sum_{j=1}^{p}\sum_{k=1}^{2}\beta_{jk}^2 = 1.$$

This becomes unwieldly if we have to check, say every degree 2 monomoial X_iX_j, so need something more efficient

Kernels - "There is no plane"

$$y_i f(x_i) \geq M(1 - \epsilon_i)$$

- Main idea: we can change up $f(x_i)$ to be nonlinear
- In fact, we do not need absolute coordinates of x_i. just need to know how close or similar to each other (draw points w/o coordinates)
- turns out just need to know general similarity between every two points - more computationally efficient
- Kernel: a more generalized way of thinking about how close or similar two samples are
- draw some examples in 1D

$f(x_i)$ should give you some more general way of talking about how far is point x_i away from the decision boundary.

- alternatively, we can simply write $f(K(x_i, x1), K(x_i, x2), \cdots, K(x_i, x_p))$
- this is essentially in relative coordinates

The kernel - replacing global coordinate with local similarity

$$K(x_i, x_i')$$

- $K(x_i, x_i')$ can be thought of as similarity function.
- An example is inner product $\langle x, x_i \rangle$ will be talked about in more detail on the next slide.
- $\langle x, x_i \rangle$ is known as a linear kernel

$$y_i f(x_i) \geq M(1 - \epsilon_i), \text{ where}$$

$$f(x) = \beta_0 + \sum_{j \in \mathcal{S}} \alpha_i K(x, x_j)$$

- The function defined is as above with whatever choice of K
- When the support vector classifier is combined with a non-linear kernel the resulting classifier is known as a support vector machine.

Inner products

$$\langle a, b \rangle = \sum_{i=1}^{r} a_i b_i$$

- sort of a measure of similarity between vectors, if they are the same length
- example draw $(1/2, \sqrt{3}/2)$ and $(0, 1)$
- example draw $(1, 0)$ and $(0, 1)$
- Example in slightly higher dimension $\langle (1, 2, 3), (5, 0, 2) \rangle = 5 + 0 + 6 = 11$

Quick computations

What are the following?

- $\langle (1,1),(0,3) \rangle = 0 + 3 = 3$
- $\langle (1,1),(3,2) \rangle = 3 + 2 = 5$
- $\langle (2,3),(0,3) \rangle = 0 + 9 = 9$
- $\langle (2,3),(3,2) \rangle = 6 + 6 = 12$

SVC via inner products

$$f(x) = \beta_0 + \sum_{i=1}^{n} \alpha_i \langle x, x_i \rangle$$

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i \langle x, x_i \rangle$$

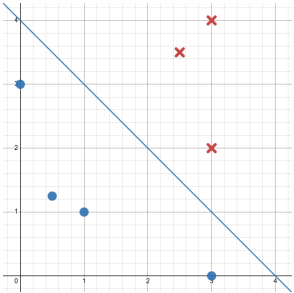
- Via some magic, there are coefficients α_i which give the linear support vector classifier
- In this notation, the x_i's are all the training points

- How to actually get it is outside the scope of this class
- To estimate the paramters $\alpha_1, \cdots, \alpha_n$ and β_0, need $\binom{n}{2} = n(n-1)/2$ inner products $\langle x_i, x_{i'} \rangle$
- BUT, α is only non-zero for support vectors. so actually few terms

Example

$$-2\sqrt{2} + \frac{\sqrt{2}}{2}X_1 + \frac{\sqrt{2}}{2}X_2 = 0$$

$$-2\sqrt{2} + \frac{\sqrt{2}}{18}\langle (X_1, X_2), (0,3)\rangle + \frac{\sqrt{2}}{6}\langle (X_1, X_2), (3,2)\rangle = 0$$



- work throught the formulation $\beta_0 + \alpha_1\langle (X_1, X_2), (0,3)\rangle + \alpha_2\langle (X_1, X_2), (3,0)\rangle + \alpha_3\langle (X_1, X_2), (3,2)\rangle = 0$ for the hyperplane
- to get unique coefficients, I let $\alpha_2 = 0$. other options are too complicated
- test $f((3,1)) = f((2,2)) = 0$ (points on hyper plane)
- test $f((1,1)) = -\sqrt{2}$
- The point: representing linear classifier $f(x)$ just needs inner products

The kernel - more generally

$$K(x_i, x_i')$$

- Swap out my inner product $\langle x, x_i \rangle$ for something potentially more complicated
- $\langle x, x_i \rangle$ is known as a linear kernel

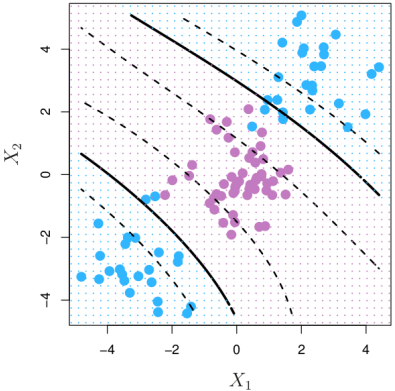
$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i)$$

- The function defined is as above with whatever choice of K
- $K(x_i, x_i')$ can be thought of as similarity function.
- When the support vector classifier is combined with a non-linear kernel the resulting classifier is known as a support vector machine.

A polynomial kernel

$$K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij} x_{i'j}\right)^d$$

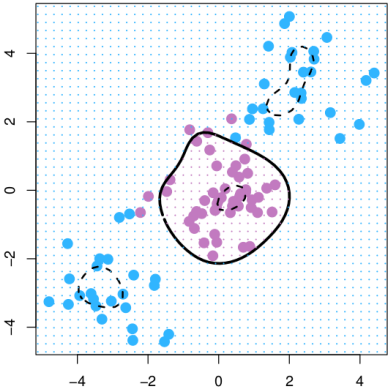
- much more flexible decision boundary.
- amounts to fitting a support vector classifier in a higher-dimensional space involving polynomials of degree d, rather than in the original feature space.
- Right: degree 3 polynomial kernel



A radial kernel

$$K(x_i, x_i') = \exp\left(-\gamma \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2\right)$$

- this is essentially a gaussian kernel
- big distance makes $\sum (x_{ij} - x_{i'j})^2$ big, but then e^{-big} is tiny.
- In $f(x)$, this means that x_i plays little role if it's far away.
- Radial kernel has local behavior



Support Vector Machine

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i)$$

- Learning function above
- Choose K in advance

Coding

Section 3

SVM with more than two classes

One-Vs-One Classification

Also called all-pairs

- Predict for $K > 2$ classes
- Construct $\binom{K}{2}$ SVMs, each of which compairs a pair of classes
- Write example with $\{$ apples, bananas, oranges, strawberries$\}$
- Assign the observation to the class which was most frequently assigned

One-Vs-All Classification

- Fit K SVMs each time comparing one class to remaining $K-1$ classes.
- Again, do { apples, bananas, oranges, strawberries}
- Figure out coefficients for that class, and determine $f_k(x) = \beta_{0k} + \beta_{1k}x_1^* + \cdots + \beta_{pk}x_p^*$
- Assign observation to the class for which $f_k(x)$ is largest

TL;DR

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i)$$

Kernels

- Linear

$$K(x_i, x_{i'}) = \sum_{j=1}^{p} x_{ij} x_{i'j}$$

- Polynomial

$$K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij} x_{i'j}\right)^d$$

- Radial

$$K(x_i, x_i') = \exp\left(-\gamma \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2\right)$$

Next time

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